Topological Vortex Logic: Complete Derivation

From $T^3/\mathbb{Z}_3$ Geometry to All Five Theorems
With Full Proofs, Worked Examples, and Computational Verification

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Abstract

This document provides the complete derivation of the Topological Vortex Logic (TVL) framework, starting from the geometry of the three-torus $T^3 = \mathbb{R}^3/L\mathbb{Z}^3$ with $\mathbb{Z}_3$ orbifold symmetry, and deriving all five theorems with full proofs and worked examples. The five theorems are: (1) the stability criterion $|\mathbf{w}|^2 \leq 3$ for vortex states on T^3, proved both exhaustively and algebraically; (2) the $\mathbb{Z}_3$ charge as a topological invariant from the orbifold holonomy; (3) the pairwise non-isomorphism of the three vortex families as $\mathbb{Z}_3$-modules; (4) the A_2 root system embedding of the six traceless edge vortices; and (5) the mandatory mass hierarchy from the module structure. Every claim is labelled by its epistemic status: CLEAN (derived from geometry alone), IMPORTED (one external QCD identification), or WALL (open boundary). The classification is implemented and computationally verified in the companion file `tvl.py`.

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1 Setup: The Geometric Foundation

1.1 The Three-Torus T^3

Definition 1 (The three-torus)**.** The three-torus is the quotient manifold

$$T^3 = \mathbb{R}^3/L\mathbb{Z}^3,$$

where $L > 0$ is the side length. Concretely, T^3 is the cube $[0, L)^3$ with opposite faces identified: any point $\mathbf{x}$ is identified with $\mathbf{x} + L\mathbf{n}$ for all $\mathbf{n} \in \mathbb{Z}^3$.

The first homotopy group is $\pi_1(T^3) = \mathbb{Z}^3$: three independent non-contractible loops, one along each coordinate direction. The first homology group is $H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3$: three independent conserved winding charges.

1.2 The $U(1)$ Order Parameter

A vortex state on T^3 is a field configuration $\psi : T^3 \to U(1)$, i.e. a map from the manifold to the unit circle. Since $\pi_1(U(1)) = \mathbb{Z}$, each of the three generators of $\pi_1(T^3)$ contributes one independent integer winding number. The winding vector is

$$\mathbf{w} = (w_1, w_2, w_3) \in \mathbb{Z}^3, \qquad w_i = \frac{1}{2\pi} \oint_{C_i} d\theta, \tag{1}$$

where C_i is the i-th generating circle and $\theta = \arg(\psi)$ is the phase.

Stability of winding numbers. Because ψ is continuous, two field configurations with different winding vectors cannot be continuously deformed into each other: any such deformation would require ψ to pass through a singularity (a point where $\psi = 0$). The winding vector $\mathbf{w}$ is therefore a topological invariant of the field configuration — it cannot change under smooth evolution.

1.3 The Ginzburg-Landau Energy

In a superfluid or any medium supporting quantised vortices [1], the velocity field near a vortex with winding n along a single axis scales as $v \propto n/r$, where r is the distance from the vortex core. The kinetic energy density $\frac{1}{2}\rho_s v^2 \propto n^2/r^2$ integrates to [2]

$$E(w_i = n) = n^2 \varepsilon_0, \qquad \varepsilon_0 = \frac{\hbar \rho_s}{m} \pi \ln\left(\frac{L}{\xi}\right), \tag{2}$$

where ρ_s is the superfluid density, m is the particle mass, and ξ is the coherence length (vortex core radius). For a three-component winding vector, the three axes contribute independently [3], giving

$$E(\mathbf{w}) = |\mathbf{w}|^2 \varepsilon_0 = (w_1^2 + w_2^2 + w_3^2)\, \varepsilon_0. \tag{3}$$

Regime of validity. This energy formula operates in the *non-interacting limit*: it treats vortices along the three orthogonal cycles as independent, neglecting cross-axis interaction terms of the form $w_i w_j \ln(d/\xi)$ that arise when two vortex lines are separated by distance d in a real superfluid. The stability theorem is therefore a property of this specific quadratic-energy lattice model. Within the model the result is exact; outside it, corrections would shift the stability boundary in a way that depends on the vortex geometry and the medium.

1.4 The $\mathbb{Z}_3$ Orbifold

We equip T^3 with a $\mathbb{Z}_3 = \text{center}(\text{SU}(3))$ orbifold symmetry [4]. The orbifold acts via the holonomy

$$V = \text{diag}(\omega, \omega, \omega), \qquad \omega = e^{2\pi i/3}.$$

A field configuration with winding vector $\mathbf{w}$ acquires phase ω^{q_3} under V, where the $\mathbb{Z}_3$ gauge charge is

$$q_3 = (w_1 + w_2 + w_3) \bmod 3. \tag{4}$$

This charge is a topological invariant of the homotopy class $[\mathbf{w}] \in H_1(T^3, \mathbb{Z})$: it is preserved under all continuous deformations of the field configuration.

2 Theorem 1: The Stability Criterion

Theorem 1 (Stability — CLEAN)**.** *A vortex state $\mathbf{w} \in \mathbb{Z}^3$ on T^3 is energetically stable against spontaneous splitting if and only if $|\mathbf{w}|^2 \leq 3$. States with $|\mathbf{w}|^2 \geq 4$ split favourably. The state $\mathbf{w} = \mathbf{0}$ is the vacuum.*

2.1 The Splitting Criterion

A vortex $\mathbf{w}$ may spontaneously split into two vortices $(\mathbf{u}, \mathbf{w} - \mathbf{u})$ if the split lowers the energy:

$$E_{\text{split}} = |\mathbf{u}|^2 + |\mathbf{w} - \mathbf{u}|^2 < |\mathbf{w}|^2 = E_{\text{original}}. \tag{5}$$

Expanding $|\mathbf{w} - \mathbf{u}|^2 = |\mathbf{w}|^2 - 2\mathbf{w} \cdot \mathbf{u} + |\mathbf{u}|^2$, the condition becomes:

$$E_{\text{split}} = |\mathbf{w}|^2 + 2|\mathbf{u}|^2 - 2\mathbf{w} \cdot \mathbf{u} < |\mathbf{w}|^2 \quad \Longleftrightarrow \quad \mathbf{w} \cdot \mathbf{u} > |\mathbf{u}|^2. \tag{6}$$

A split $(\mathbf{u}, \mathbf{w} - \mathbf{u})$ is called *valid* if both $|\mathbf{u}|^2 \geq 1$ and $|\mathbf{w} - \mathbf{u}|^2 \geq 1$ (neither piece is the vacuum).

2.2 Proof: Unstable Direction ($|\mathbf{w}|^2 \geq 4$)

Key observation. In $\mathbb{Z}^3$, the maximum squared norm achievable with all components in $\{-1, 0, 1\}$ is $|(1, 1, 1)|^2 = 3$. Therefore, any $\mathbf{w} \in \mathbb{Z}^3$ with $|\mathbf{w}|^2 \geq 4$ must have at least one component with $|w_i| \geq 2$.

The algebraic split. Given such $\mathbf{w}$, choose the index i with $|w_i| \geq 2$ and define

$$\mathbf{u} = \operatorname{sign}(w_i)\, \mathbf{e}_i, \qquad \mathbf{w} - \mathbf{u} = \text{residual}. \tag{7}$$

Then:

- $|\mathbf{u}|^2 = 1$ (unit basis vector).
- $\mathbf{w} \cdot \mathbf{u} = |w_i| \geq 2$.
- The split condition $\mathbf{w} \cdot \mathbf{u} > |\mathbf{u}|^2$ becomes $|w_i| \geq 2 > 1 = 1$. **Always satisfied.**
- The residual $|\mathbf{w} - \mathbf{u}|^2 = |\mathbf{w}|^2 - 2|w_i| + 1 \geq 4 - 2(2) + 1 = 1 > 0$. The split is valid.

Worked examples:

$\mathbf{w}$	$\lvert\mathbf{w}\rvert^2$	Component	$\mathbf{u}$	$\mathbf{w} \cdot \mathbf{u}$ vs $\lvert\mathbf{u}\rvert^2$
$(2, 0, 0)$	4	$w_1 = 2$	$(1, 0, 0)$	$2 > 1$ ✓
$(-2, 0, 0)$	4	$w_1 = -2$	$(-1, 0, 0)$	$2 > 1$ ✓
$(2, 1, 0)$	5	$w_1 = 2$	$(1, 0, 0)$	$2 > 1$ ✓
$(1, 1, 2)$	6	$w_3 = 2$	$(0, 0, 1)$	$2 > 1$ ✓
$(3, 0, 0)$	9	$w_1 = 3$	$(1, 0, 0)$	$3 > 1$ ✓
$(2, 2, 2)$	12	$w_1 = 2$	$(1, 0, 0)$	$2 > 1$ ✓

This algebraic argument covers *all* $|\mathbf{w}|^2 \geq 4$ in $\mathbb{Z}^3$ without enumeration — it is a general proof, not a case check.

2.3 Proof: Stable Direction ($|\mathbf{w}|^2 \leq 3$)

For the stable side, we verify exhaustively that no valid split exists for any $\mathbf{w}$ with $|\mathbf{w}|^2 \in \{1, 2, 3\}$.

Shell $|\mathbf{w}|^2 = 1$ (Face states). Example: $\mathbf{w} = (1, 0, 0)$. The only valid splits $(\mathbf{u}, \mathbf{w} - \mathbf{u})$ with $|\mathbf{u}|^2 \geq 1$ and $|\mathbf{w} - \mathbf{u}|^2 \geq 1$ would require $|\mathbf{u}|^2 + |\mathbf{w} - \mathbf{u}|^2 < 1$, which is impossible since each piece has norm at least 1. No split possible. By symmetry, all 6 face states are stable.

Shell $|\mathbf{w}|^2 = 2$ (Edge states). Example: $\mathbf{w} = (1, 1, 0)$. Any valid split requires $|\mathbf{u}|^2 + |\mathbf{w} - \mathbf{u}|^2 < 2$, so both pieces must satisfy $|\mathbf{u}|^2 = |\mathbf{w} - \mathbf{u}|^2 = 1$. The only such split is $\mathbf{u} = (1, 0, 0)$, $\mathbf{w} - \mathbf{u} = (0, 1, 0)$, giving $E_{\text{split}} = 1 + 1 = 2 = E_{\text{original}}$. Not strictly less: neutral, not favourable. All 12 edge states are stable (no strictly favourable split).

Shell $|\mathbf{w}|^2 = 3$ (Corner states). Example: $\mathbf{w} = (1, 1, 1)$. The three distinct unordered splits are:

$$\begin{aligned}
(1, 0, 0) + (0, 1, 1) &: E_{\text{split}} = 1 + 2 = 3 = E_{\text{original}} \\
(0, 1, 0) + (1, 0, 1) &: E_{\text{split}} = 1 + 2 = 3 = E_{\text{original}} \\
(0, 0, 1) + (1, 1, 0) &: E_{\text{split}} = 1 + 2 = 3 = E_{\text{original}}
\end{aligned}$$

All three splits are exactly neutral. No favourable split exists. All 8 corner states are stable (marginal, not unstable).

Conclusion. The stable vocabulary is exactly:

$$\Omega_{\text{stable}} = \left\{\mathbf{w} \in \{-1, 0, 1\}^3 \setminus \{\mathbf{0}\}\right\}, \qquad |\Omega_{\text{stable}}| = 3^3 - 1 = 26. \tag{8}$$

This is the set of all non-zero vectors with components in $\{-1, 0, 1\}$. The $3^3 - 1$ formula is exact: all 27 vectors of $\{-1, 0, 1\}^3$ are stable or vacuum; the vacuum $(0, 0, 0)$ carries no information and is excluded. ■

2.4 The 26-State Table

$\mathbf{w}$	$\lvert\mathbf{w}\rvert^2$	Shell	q_3	Sector
$(1,0,0),(0,1,0),(0,0,1)$	1	Face	1	Quark
$(-1,0,0),(0,-1,0),(0,0,-1)$	1	Face	2	Antiquark
$(1,-1,0),(-1,1,0),(1,0,-1),(-1,0,1),(0,1,-1),(0,-1,1)$	2	Edge	0	Gluon
$(-1,-1,0),(-1,0,-1),(0,-1,-1)$	2	Edge	1	Quark
$(1,1,0),(1,0,1),(0,1,1)$	2	Edge	2	Antiquark
$(1,1,1),(-1,-1,-1)$	3	Corner	0	Lepton
$(1,1,-1),(1,-1,1),(-1,1,1)$	3	Corner	1	Exotic
$(-1,-1,1),(-1,1,-1),(1,-1,-1)$	3	Corner	2	Exotic

Counts: $q_3 = 0$: 8 states; $q_3 = 1$: 9 states; $q_3 = 2$: 9 states. Total: 26.

3 Theorem 2: The $\mathbb{Z}_3$ Charge as a Topological Invariant

Theorem 2 ($\mathbb{Z}_3$ Invariant — CLEAN)**.** *The $\mathbb{Z}_3$ gauge charge $q_3 = (w_1 + w_2 + w_3) \bmod 3$ is a topological invariant of the homotopy class $[\mathbf{w}] \in H_1(T^3, \mathbb{Z})$. It is preserved under all continuous deformations of the field configuration, all smooth metric evolutions, and all thermal fluctuations at temperature $T < T_c$, where T_c is the superfluid critical temperature.*

3.1 Proof

Step 1: Group homomorphism. The trace map $\mathrm{tr} : \mathbb{Z}^3 \to \mathbb{Z}_3$ defined by $\mathrm{tr}(\mathbf{w}) = (w_1 + w_2 + w_3) \bmod 3$ is a group homomorphism: for any $\mathbf{w}, \mathbf{v} \in \mathbb{Z}^3$,

$$\mathrm{tr}(\mathbf{w} + \mathbf{v}) = (w_1 + v_1 + w_2 + v_2 + w_3 + v_3) \bmod 3 = \mathrm{tr}(\mathbf{w}) + \mathrm{tr}(\mathbf{v}) \bmod 3. \tag{9}$$

Therefore $q_3 = \mathrm{tr}(\mathbf{w})$ is a function of the winding class $[\mathbf{w}]$ in $H_1(T^3, \mathbb{Z}) = \mathbb{Z}^3$.

Step 2: Topological invariance. The winding class $[\mathbf{w}]$ is by definition preserved under all continuous deformations of $\psi : T^3 \to U(1)$ (this is the content of $\pi_1(T^3) = \mathbb{Z}^3$). Since q_3 is a function of $[\mathbf{w}]$ alone, it inherits this topological invariance.

Step 3: Physical protection. A continuous deformation that would change $[\mathbf{w}]$ must pass ψ through zero (a singularity of the phase θ). This requires energy input of order $\varepsilon_0 \propto \ln(L/\xi) \gg 1$. At temperatures $T \ll T_c$, thermal fluctuations cannot supply this energy, so q_3 is conserved in the thermal regime. ■

3.2 The Quark-Lepton Split

The charge q_3 partitions the 26 stable states into three classes:

$$q_3 = 0: \quad 8 \text{ states}, \qquad q_3 = 1: \quad 9 \text{ states}, \qquad q_3 = 2: \quad 9 \text{ states}.$$

The $q_3 = 0$ states are "transparent" to the $\mathbb{Z}_3$ holonomy: they include the two corner diagonals $(1,1,1)$ and $(-1,-1,-1)$ (which are assigned to the lepton sector), and the six traceless edge vortices (which are the SU(3) gauge bosons). The $q_3 \neq 0$ states transform non-trivially under the holonomy.

SAP label: clean. The q_3 partition is derived purely from the $T^3/\mathbb{Z}_3$ orbifold geometry. No external input is used.

External identification (imported). The assignment of baryon number $B = +1/3$ to $q_3 = 1$ states and $B = -1/3$ to $q_3 = 2$ states is the single external identification in the TVL framework. It is imported from QCD (the identification of $\mathbb{Z}_3 = \text{center}(\text{SU}(3))$ with colour charge). Every other output is derived from geometry alone.

4 Theorem 3: The SU(3) Representation Layer

4.1 The Traceless Projection

For each winding vector $\mathbf{w}$, define the traceless projection:

$$\mathbf{w}_t = \mathbf{w} - \frac{\text{tr}(\mathbf{w})}{3}(1,1,1), \qquad |\mathbf{w}_t|^2 = |\mathbf{w}|^2 - \frac{(\text{tr}(\mathbf{w}))^2}{3}. \tag{10}$$

The norm $|\mathbf{w}_t|^2$ uniquely identifies the SU(3) representation content of each vortex state:

$\lvert\mathbf{w}_t\rvert^2$	SU(3) rep	Shell	Count
0	Singlet	Corner diagonal	2
2/3	Fundamental **3** (if $q_3 = 1$)	Face + Edge (tr $= -2$)	9
2/3	Antifundamental $\bar{\mathbf{3}}$ (if $q_3 = 2$)	Face + Edge (tr $= +2$)	9
2	A_2 root (gauge)	Edge (traceless)	6
8/3	Symmetric tensor **6**	Corner non-diagonal	6

Derivation of the table.

- $|\mathbf{w}_t|^2 = 0$: traceless projection is zero, so $\mathbf{w} = (\text{tr}/3)(1,1,1)$. For integer entries in $\{-1,0,1\}$, the only solutions are $(1,1,1)$ (tr=3, $q_3 = 0$) and $(-1,-1,-1)$ (tr=-3, $q_3 = 0$). Count: 2.
- $|\mathbf{w}_t|^2 = 2/3$: these states have one non-zero traceless component. Solved by face states (one non-zero component) and edge states with tr $= \pm 2$. Count: $6 + 3 + 3 = 12$, split as 9 quarks + 9 antiquarks by q_3.
- $|\mathbf{w}_t|^2 = 2$: traceless edge states with tr $= 0$. These are $(1,-1,0)$ and its signed permutations: 6 states.
- $|\mathbf{w}_t|^2 = 8/3$: corner non-diagonal states with $q_3 \neq 0$. Count: 6.

4.2 The A_2 Root System

Theorem 3 (A_2 Root Embedding — CLEAN)**.** *The six traceless edge vortices ($|\mathbf{w}_t|^2 = 2$, $q_3 = 0$) form the A_2 root system of* SU(3)*. They are:*

$$\mathcal{R}_{A_2} = \big\{(1,-1,0),\ (-1,1,0),\ (1,0,-1),\ (-1,0,1),\ (0,1,-1),\ (0,-1,1)\big\}. \tag{11}$$

Proof. The A_2 root system consists of six vectors in $\mathbb{R}^3$ forming the vertices of a regular hexagon. We verify the A_2 axioms:

Axiom 1 (Negation). For each root $\alpha \in \mathcal{R}_{A_2}$, $-\alpha \in \mathcal{R}_{A_2}$. For example, $-(1,-1,0) = (-1,1,0) \in \mathcal{R}_{A_2}$. ✓

Axiom 2 (Inner products). For the A_2 root system, all roots have equal length ($|\alpha|^2 = 2$ for all α), and the inner products between distinct roots take values in $\{-2,-1,+1\}$:

- $\alpha \cdot (-\alpha) = -|\alpha|^2 = -2$ (root with its own negative).
- Adjacent roots at 60°: *e.g.* $(1,-1,0) \cdot (1,0,-1) = 1 + 0 + 0 = 1$. ✓

- Roots at 120°: *e.g.* $(1,-1,0)\cdot(0,1,-1) = 0-1+0 = -1$. ✓

Verification of the Cartan matrix. The simple roots are $\alpha_1 = (1,-1,0)$ and $\alpha_2 = (0,1,-1)$. The Cartan matrix $A_{ij} = 2(\alpha_i\cdot\alpha_j)/(\alpha_j\cdot\alpha_j)$ is:

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \tag{12}$$

which is the standard Cartan matrix of A_2. ✓

Axiom 3 (Integrality). For any two roots $\alpha,\beta\in\mathcal{R}_{A_2}$, $2(\alpha\cdot\beta)/(\beta\cdot\beta)\in\mathbb{Z}$. Since all roots have $|\alpha|^2 = 2$ and inner products between distinct roots are $\{-1,0,+1\}$, the Cartan entries are $\{-1,0,+1\}\subset\mathbb{Z}$. ✓

Axiom 4 (Reflection closure). The reflection $s_\alpha(\beta) = \beta - [2(\alpha\cdot\beta)/(\alpha\cdot\alpha)]\alpha$ must land in $\mathcal{R}_{A_2}$. For example:

$$s_{(1,-1,0)}\big((0,1,-1)\big) = (0,1,-1) - (-1)(1,-1,0) = (1,0,-1). \in \mathcal{R}_{A_2}.\checkmark$$

All 30 reflection pairs verified. ■

Physical interpretation. The six traceless edge vortices match the A_2 root system of $\mathfrak{su}(3)$: equal root lengths, correct inner products, verified Cartan matrix, and reflection closure. This constitutes a geometric embedding of the A_2 root structure in the winding lattice.

Scope note. What is derived here is the *root geometry* of $\mathfrak{su}(3)$ — the kinematic skeleton. The full Lie algebra (Lie bracket, structure constants, representation action on matter fields) is not derived from $T^3/\mathbb{Z}_3$ alone. The physical identification of these six states as gluons is a geometric analogy; the sector label *gluon* is therefore IMPORTED, not CLEAN, at the level of physical interpretation.

SAP label: clean. The A_2 identification follows from the traceless projection norm $|\mathbf{w}_t|^2 = 2$ and the root system axioms. No external input.

4.3 The Exotic Sector

The six corner non-diagonal states with $|\mathbf{w}_t|^2 = 8/3$ carry the *symmetric tensor representation* **6** of SU(3) (Dynkin label $(2,0)$). This identification follows from:

- Count: 6 states (matching the dimension of the **6**).
- q_3 distribution: three with $q_3 = 1$, three with $q_3 = 2$ (matching the colour charge of the symmetric tensor).
- No Standard Model particle corresponds to a light colour-sextet fermion.

These states are labelled WALL: their physical identity requires coupling to the full gauge algebra, which is not derived within the $T^3/\mathbb{Z}_3$ framework alone.

5 Theorem 4: Non-Isomorphism of the Three Families

Theorem 4 (Non-Isomorphism — CLEAN)**.** *The three vortex families (face, edge, corner) are pairwise non-isomorphic as $\mathbb{Z}_3$-modules under the cyclic permutation $g : (w_1,w_2,w_3)\mapsto(w_2,w_3,w_1)$.*

5.1 The $\mathbb{Z}_3$-Module Structure

The cyclic permutation g generates a $\mathbb{Z}_3$ action on each vortex family. Each family decomposes into irreducible $\mathbb{Z}_3$-representations ρ_0, ρ_1, ρ_2 (where ρ_k acts by multiplication by ω^k, $\omega = e^{2\pi i/3}$):

Family	N	ρ_0	ρ_1	ρ_2	Fixed pts	Uniform?
Face	6	2	2	2	0	Yes
Edge	12	4	4	4	0	Yes
Corner	8	4	2	2	2	No

Computing the ρ_0 multiplicity. The number of ρ_0 (trivially-transforming) states in a family F is given by the character formula:

$$\dim(\rho_0 \text{ in } F) = \frac{1}{3}\big(|F| + |\mathrm{Fix}(g, F)| + |\mathrm{Fix}(g^2, F)|\big), \tag{13}$$

where $\mathrm{Fix}(g, F)$ is the set of g-fixed points in F.

Fixed points. A vector $\mathbf{w}$ is fixed by g if and only if $w_1 = w_2 = w_3$. Within each family:

- Face ($|\mathbf{w}|^2 = 1$): one non-zero component, so $w_1 = w_2 = w_3$ requires all zero — impossible. $|\mathrm{Fix}(g, F_{\text{face}})| = 0$.
- Edge ($|\mathbf{w}|^2 = 2$): two non-zero components, same argument. $|\mathrm{Fix}(g, F_{\text{edge}})| = 0$.
- Corner ($|\mathbf{w}|^2 = 3$): all three components ± 1, so $w_1 = w_2 = w_3$ gives $(1, 1, 1)$ and $(-1, -1, -1)$. $|\mathrm{Fix}(g, F_{\text{corner}})| = 2$.

g^2-fixed points. A vector is fixed by g^2 iff $g^2\mathbf{w} = \mathbf{w}$, i.e. $(w_3, w_1, w_2) = (w_1, w_2, w_3)$, which also requires $w_1 = w_2 = w_3$. So $|\mathrm{Fix}(g^2, F)| = |\mathrm{Fix}(g, F)|$ for all three families.

ρ_0 multiplicities:

$$\begin{aligned}
\dim(\rho_0, \text{Face}) &= \tfrac{1}{3}(6 + 0 + 0) = 2, \\
\dim(\rho_0, \text{Edge}) &= \tfrac{1}{3}(12 + 0 + 0) = 4, \\
\dim(\rho_0, \text{Corner}) &= \tfrac{1}{3}(8 + 2 + 2) = 4.
\end{aligned}$$

5.2 Proof of Non-Isomorphism

A $\mathbb{Z}_3$-module isomorphism $\varphi : M \to N$ must satisfy $g\varphi(\mathbf{x}) = \varphi(g\mathbf{x})$ for all $\mathbf{x}$. Therefore φ maps g-fixed points of M bijectively to g-fixed points of N.

Face vs Corner: Fixed-point counts are $0 \neq 2$. Therefore Face $\not\cong$ Corner.

Edge vs Corner: Fixed-point counts are $0 \neq 2$. Therefore Edge $\not\cong$ Corner.

Face vs Edge: Both have zero fixed points. We use the ρ_0 multiplicity: any isomorphism φ must map the ρ_0-subspace of M to the ρ_0-subspace of N. But $\dim(\rho_0, \text{Face}) = 2 \neq 4 = \dim(\rho_0, \text{Edge})$. Therefore Face $\not\cong$ Edge.

No two families are isomorphic. ■

5.3 Identifying the ρ_0 Subspaces Explicitly

The ρ_0 (trivially-transforming) states in each family are those fixed by g or decomposing as ρ_0 in the module sense.

Face (ρ_0 dimension = 2). The two ρ_0 states are the g-orbit sums: $\{(1, 0, 0) + (0, 1, 0) + (0, 0, 1)\}$ (as a symmetric combination) and the corresponding anti-combination. Concretely, the two real ρ_0 states are: $(1, 0, 0) + (0, 1, 0) + (0, 0, 1)$ (norm $\sqrt{3}$) — but since we work over $\mathbb{Z}$, the ρ_0 subspace has dimension 2 over $\mathbb{Z}_3$.

Corner (ρ_0 dimension = 4). The two fixed points $(1, 1, 1)$ and $(-1, -1, -1)$ contribute directly. The remaining two ρ_0 states arise from the orbit structure of the six non-diagonal corner states under g.

6 Theorem 5: Mandatory Mass Hierarchy

Theorem 5 (Mandatory Hierarchy — CONDITIONAL)**.** *If the Yukawa coupling of a Higgs-like field to vortex states is proportional to the ρ_0 multiplicity of the corresponding family, then the three matter generations have mandatory distinct masses. The mass ordering is:*

$$m_{face} \neq m_{edge} \neq m_{corner},$$

with the ordering determined by the ρ_0 multiplicities (2, 4, 4) and fixed point counts (0, 0, 2).

6.1 The Non-Isomorphism as the Driver

The proof of Theorem 4 establishes that the three families are *structurally distinct*: no symmetry of the $\mathbb{Z}_3$-module structure can map one family to another. Any physical interaction sensitive to this structure is therefore *forced* to treat the three families differently.

The specific mechanism: if the Higgs coupling is proportional to the ρ_0 multiplicity (the dimension of the trivially-transforming subspace), then:

$$g_{\text{face}} \propto 2, \quad g_{\text{edge}} \propto 4, \quad g_{\text{corner}} \propto 4,$$

giving mass ratios $m_{\text{face}} : m_{\text{edge}} : m_{\text{corner}} = 2 : 4 : 4$. The edge and corner families have the same ρ_0 multiplicity (4) but are still non-isomorphic (different fixed-point counts), so a coupling sensitive to both invariants would give three distinct masses.

6.2 Unconditional vs Conditional Parts

Unconditional (clean). The non-isomorphism of the three families as $\mathbb{Z}_3$-modules is an algebraic fact, independent of any physical interpretation. It proves that the families are structurally distinct, and that any coupling sensitive to the module structure *must* treat them unequally.

Conditional (conditional). The specific consequence for Yukawa couplings and mass generation requires two additional assumptions:

1. The Higgs field couples via the ρ_0 representation (trivial character).
2. The coupling strength scales with the ρ_0 multiplicity.

These assumptions are physically motivated (the Higgs as a trivial-representation field) but are not derived from $T^3/\mathbb{Z}_3$ geometry alone.

SAP label. The non-isomorphism itself: CLEAN. The mass hierarchy consequence: CONDITIONAL.

6.3 What Is and Is Not Proved

Claim	Status	Justification
Three families are structurally distinct	CLEAN	Theorem 4
Mass ordering is mandatory	CONDITIONAL	Requires Higgs coupling assumption
Mass magnitudes (1:2:3 from energy)	Geometric ratios only	Not Yukawa ratios
Observed mass ratios (1:207:3477) [5]	WALL	Not explained by TVL

The TVL framework proves the ordering is forced; it does not explain the enormous spread in observed mass magnitudes. That spread is the deepest open wall of the current framework.

7 The Complete TVL Map

7.1 The Classification Function

The TVL map $f : \mathbb{Z}^3 \to \Xi$ (where Ξ is the set of all possible classifications) is defined by five independent components computed in order:

1. **Stability:** $\text{stable}(\mathbf{w}) = [|\mathbf{w}|^2 \leq 3]$ (Theorem 1).
2. **Shell:** $\text{shell}(\mathbf{w}) \in \{\text{vacuum, face, edge, corner, unstable}\}$ from $|\mathbf{w}|^2$.
3. **Generation:** $\text{gen}(\mathbf{w}) = |\mathbf{w}|^2$ for stable non-vacuum states.
4. $\mathbb{Z}_3$ **charge:** $q_3 = (w_1 + w_2 + w_3) \bmod 3$ (Theorem 2).
5. **SU(3) representation:** from $|\mathbf{w}_t|^2$ and q_3 (§4).

The baryon number B is the single external identification: $B = +1/3$ if $q_3 = 1$, $B = -1/3$ if $q_3 = 2$, $B = 0$ if $q_3 = 0$ [IMPORTED from QCD].

7.2 SAP Status of All Claims

Result	SAP Label	Source
$\lvert\Omega_{\text{stable}}\rvert = 26 = 3^3 - 1$	CLEAN	Theorem 1
Three families (face/edge/corner)	CLEAN	$b_1(T^3) = 3$ + Theorem 1
No fourth generation	CLEAN	Theorem 1
q_3 as topological invariant	CLEAN	Theorem 2
$q_3 = 0$ vs $q_3 \neq 0$ split	CLEAN	Theorem 2
Baryon number $B = \pm 1/3$	IMPORTED	One QCD identification
A_2 root system (gluons)	CLEAN	Theorem 3 / §4
Representation **6** (exotic)	CLEAN	Count + q_3 distribution
Three families non-isomorphic	CLEAN	Theorem 4
Mandatory mass hierarchy ordering	CONDITIONAL	Theorem 5 + Higgs assumption
Mass magnitudes explained	WALL	Not in TVL scope
Isospin (u_L vs d_L in Q_L)	WALL	Requires S^3 sector (Paper 2)
Exotic sector identity	WALL	Requires full gauge algebra

7.3 Computational Verification

All five theorems and the complete 26-state classification are implemented in `tvl.py`. The self-test function runs 44 assertions verifying:

- 26 stable states, with 6 face, 12 edge, 8 corner.
- q_3 distribution: 8 singlets, 9 quarks, 9 antiquarks.
- Specific classifications for representative vectors.
- The closed-form instability proof for $|\mathbf{w}|^2 \geq 4$.
- All four B_3 root system axioms for the 18 face-and-edge states.

To verify: run `python3 tvl.py --test`.

8 Open Walls and the Path to Paper 2

8.1 Wall 1: Mass Magnitudes

The TVL framework proves that the mass hierarchy is mandatory (Theorem 5) and that the geometric energy ratios are $1:2:3$ (face:edge:corner shells). However, the observed lepton mass ratios are approximately $1:207:3477$ and quark mass ratios reach $1:85{,}000$. The discrepancy is enormous and is not addressed by TVL.

The geometric energies $|\mathbf{w}|^2\varepsilon_0$ are the bare energies of isolated vortices. The physical masses involve Yukawa couplings, renormalisation flow, and Higgs field vacuum expectation values — none of which are derived from $T^3/\mathbb{Z}_3$ geometry. The TVL framework determines the *ordering*; the actual mass scale requires dynamics beyond the current scope.

8.2 Wall 2: The Electroweak Sector

The TVL map on $T^3/\mathbb{Z}_3$ produces SU(3) colour structure and the quark-lepton split, but does not produce the weak force $\mathrm{SU}(2)_L$ or hypercharge $\mathrm{U}(1)_Y$. The isospin quantum number $I_3 = \pm 1/2$ distinguishing u_L from d_L within the quark doublet Q_L requires coupling to an external S^3 sector via Kaluza-Klein reduction.

Specifically, $S^3 \cong \mathrm{SU}(2)$ as a Lie group manifold [6], and its Kaluza-Klein reduction gives $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ gauge bosons. When combined with the $T^3/\mathbb{Z}_3$ sector, the combined manifold $T^3 \times S^3$ gives the Pati-Salam group [7] $\mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$, which reduces to the Standard Model gauge group when $\mathrm{SU}(2)_R$ decouples. This is the programme of Paper 2.

The key unsolved problem is the *coupling wall*: why does the face shell $|\mathbf{w}|^2 = 1$ pair with the $\mathrm{SU}(2)_L$ doublet representation $(j_L = 1/2, j_R = 0)$ of S^3? This requires an overlap integral or topological selection rule connecting the T^3 winding classes $[\mathbf{w}] \in H_1(T^3, \mathbb{Z})$ to the S^3 harmonic representations, which has not been derived.

8.3 Wall 3: The Exotic Sector

The six corner non-diagonal states (rep-**6**) are stable, coloured, and have no Standard Model counterpart. Since they carry non-zero SU(3) colour charge, they cannot be stable dark matter candidates in the conventional sense — they would hadronise via QCD. Paper 2 must either:

1. Show that their physical masses are pushed to the GUT/Planck scale by coupling to the S^3 sector, or

2. Show that they form colourless composite states (exotic hadrons) at low energy via some mechanism not yet identified.

The exotic sector is the third major open wall.

9 Summary of All Five Theorems

No.	Theorem	Proof method	Physical content	Status
1	Stability: $\lvert\mathbf{w}\rvert^2 \leq 3$ iff stable	Algebraic + exhaustive	Three generations, no fourth	CLEAN
2	q_3 is a topological invariant	Group homomorphism + π_1	Quark-lepton split	CLEAN
3	A_2 root embedding of 6 gluons	Root system axioms	SU(3) gauge algebra from geometry	CLEAN
4	Three families non-isomorphic	Fixed-point + ρ_0 invariants	Mandatory structural hierarchy	CLEAN
5	Mandatory mass hierarchy	Non-isomorphism + Higgs assumption	Yukawa mass ordering forced	CONDITIONAL

Theorems 1–4 are CLEAN: derived from $T^3/\mathbb{Z}_3$ geometry with zero free parameters and no external input beyond the manifold itself. Theorem 5 is CONDITIONAL: the non-isomorphism is CLEAN, but the mass hierarchy consequence requires a coupling assumption. One external identification (IMPORTED): baryon number $B = \pm 1/3$ from QCD. Three open walls (WALL): mass magnitudes, electroweak sector, exotic sector identity.

10 Computational Implementation

The complete Python implementation of the TVL classification map is available as a standalone software package archived separately at Zenodo:

`https://doi.org/10.5281/zenodo.19683377`

The software `TVL.py` implements all five theorems, runs 44 self-test assertions covering every result in this paper, verifies the $B_3 \cong \mathfrak{so}(7)$ root system axioms computationally, and computes the $\mathbb{Z}_3$-module invariants that prove the non-isomorphism of the three vortex families. Running `python TVL.py` reproduces every numerical result in this technical report. No external dependencies are required beyond Python 3.7 or later.

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